THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — MATHS · EASY LEVEL UNIT TEST SET 1

Chapters 1–4: **Real Numbers, Polynomials, Pair of Linear Equations in Two Variables, Quadratic Equations** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. Easy level: direct, single-step questions.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. The HCF of 12 and 18 is: 1

(a) 6 **(b)** 3 **(c)** 36 **(d)** 12

2. If $p(x) = x^2 - 3x + 2$, then $p(4)$ is: 1

(a) 8 **(b)** 30 **(c)** 4 **(d)** 6

3. The pair of equations $x + 2y = 3$ and $2x + 4y = 6$ has: 1

(a) infinitely many solutions (coincident lines)
(b) no solution (parallel lines)
(c) exactly two solutions
(d) exactly one solution (intersecting lines)

4. Which of the following is a quadratic equation? 1

(a) $x^2 + 3x + 2 = 0$ **(b)** $x + 3 = 2$ **(c)** $3x + 2 = 0$ **(d)** $x^3 + 3x + 2 = 0$

5. The LCM of 4 and 6 is: 1

(a) 2 **(b)** 10 **(c)** 12 **(d)** 24

6. The zero of the linear polynomial $2x - 8$ is: 1

(a) -8 **(b)** -4 **(c)** 4 **(d)** 8

7. Which of the following pairs (x, y) is a solution of $2x + 3y = 12$? 1

(a) $(3, 2)$ **(b)** $(-3, 2)$ **(c)** $(2, 3)$ **(d)** $(3, -2)$

8. Which of the following is a root of $x^2 - 5x + 6 = 0$? 1

(a) 2 **(b)** -2 **(c)** 5 **(d)** 1

9. The number of prime factors (counted with repetition) of 12 is: 1

(a) 6 **(b)** 3 **(c)** 4 **(d)** 2

10. The sum of the zeroes of $x^2 - 5x + 6$ is: 1

(a) 11 **(b)** 5 **(c)** 6 **(d)** -5

11. If $x = 2, y = 1$ is a solution of $2x + 3y = k$, then k is: 1

(a) 15 **(b)** 8 **(c)** 7 **(d)** 1

12. In $2x^2 - 3x + 5 = 0$, the value of b is: 1

(a) 2 **(b)** -3 **(c)** 3 **(d)** 5

13. Which of the following is an irrational number? 1

(a) $\sqrt{25}$ **(b)** $\sqrt{7}$ **(c)** 25 **(d)** $\frac{5}{8}$

14. The product of the zeroes of $x^2 - 5x + 6$ is: 1

(a) 5 **(b)** 6 **(c)** 11 **(d)** -6

15. The sum of two numbers is 10 and their difference is 4. The larger number is: 1

(a) 7 **(b)** 14 **(c)** 3 **(d)** 6

SECTION — B (Very Short Answer Type) — 5 × 2 = 10

Answer in about 20–30 words each.

16. Find the HCF of 18 and 24 by prime factorisation. 2

17. Find the value of $2x^2 - 3x + 1$ at $x = 3$. 2

18. Check whether the pair $2x + 3y = 7$, $4x + 6y = 14$ is consistent. 2

19. Check whether $x = 2$ is a root of $x^2 - 7x + 10 = 0$. 2

20. Express 140 as a product of prime factors. 2

SECTION — C (Short Answer Type) — 5 × 3 = 15

Answer in about 40–60 words each.

21. Find the LCM and HCF of 26 and 91 by prime factorisation. 3

22. Find the zeroes of $x^2 - 8x + 15$ by splitting the middle term and verify the sum of zeroes. 3

23. Solve by elimination: $x + y = 8$ and $x - y = 2$. 3

24. Solve by factorisation: $x^2 - 5x + 6 = 0$. 3

OR

Verify that HCF $\times$ LCM $=$ product of the numbers for 12 and 30. 3

25. Find a quadratic polynomial whose sum and product of zeroes are 5 and 6 respectively. 3

OR

Solve by the substitution method: $2x + 3y = 18$ and $x + y = 7$. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

26. Find the HCF and LCM of 135 and 225 by prime factorisation and verify that HCF $\times$ LCM $=$ product. 5

OR

Find the zeroes of $x^2 - 5x + 6$ and verify the relationship between the zeroes and the coefficients. 5

27. Solve by elimination: $2x + 3y = 12$ and $5x + 4y = 23$. 5

OR

Solve by factorisation: $2x^2 - 7x + 3 = 0$. 5

END OF QUESTION PAPER — **Total: 50 Marks**

MASTERMIND